# Asset CIA Analysis (10 minutes)

| Asset | Confidentiality (1-5) | Integrity (1-5) | Availability (1-5) | Justification<br>*Why did you rate it this way?* |
|---|---|---|---|---|
| Pressure/flow readings | 5 | 5 | 5 | 1.Revealing Data exposes vital information about the system and its usage.(C)<br>2.incorrect Reading could hide a dangerous condition causing a water burst.(I)<br>3.The readings are needed for monitoring and detecting dangerous conditions.(A) |
| Gate control commands | 3 | 5 | 5 | 1.Unauthorized access could reveal information about how the gates are operated.(C)<br>2.Altered commands could open/close gates incorrectly leading to serious water congestions/risks.(I)<br>3.Loss of control access prevents operators from safely managing the gates when needed.(A) |
| Emergency shutoff capability | 4 | 5 | 5 | 1.Information about the emergency system potentially helps the attacker in understanding the safety controls.(C)<br>2.Altered shutoff commands prevents an emergency response or trigger a shutdown.(I)<br>3.The system must be available during an emergency to stop the flow and prevent further damage.(A) |
| Dashboard login credentials | 5 | 5 | 5 | 1.Credentials are critical.Exposure could allow unauthorized access to the entire system.(C)<br>2.Compromised credentials could allow attackers to modify commands,readings and configurations.(I)<br>3.Authorized users cannot access the dashboard then they may be unable to monitor and control the system.(A) |

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| Consumption/savings data | 4 | 3 | 3 | 1.Exposure of this data can reveal operational patterns and sensitive information.(C)<br>2.Incorrect Data leads to inaccurate reports,financial decisions and calculations,might affect stance with a rival firm.(I)<br>3.Temporary loss might not immediately create a safety emergency.(A) |
| Leak detection alerts | 3 | 5 | 5 | 1.Alerts may reveal information about system conditions and incidents.(C)<br>2.False or modified alerts cause the operators to miss a real leak or create a real leak.(I)<br>3.If alerts aren't available,a leak may go undetected causing major safety and financial consequences. |

# Attack Mapping

For each attack type you learned, identify **one specific scenario** at The Grand Marina:

| Attack Type | Specific Scenario | Which Asset Is Targeted | CIA Impact |
|---|---|---|---|
| Eavesdropping | An attacker secretly listens to network traffic and captures consumption or communication data . | 1.Consumption/ savings<br>2.Gate command control | **Confidentiality** sensitive operational information could be exposed. |
| Spoofing | An attacker pretends to be a legitimate sensor and sends false pressure reading to the system | 1.Pressure flow reading<br>2.Leak detection alert | **Integrity** False readings could hide an underlying pressure condition and contribute to burst Water pipe. |

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|---|---|---|---|
| Replay Attack | An attacker intercepts and captures gate command and sends it again to make the gate perform the same action | Gate control commands | **Integrity**<br>Old commands could be used to cause unauthorized or unsafe gate operations. |
| Man-in-the-Middle | An attacker intercepts communications between the dashboard and the IoT system,modifies the shutoff command before it reaches the device | Emergency shutoff capability | **Integrity**<br>The command could be altered or prevented,delaying an emergency response. |
| Denial of Service | An attacker floods the IoT network So leak detection alerts never reach the dashboard, OR floods the dashboard with traffic making it difficult for the operator to use the dashboard and to monitor/control. | 1.Leak detection alerts<br>2.Dashboard | **Availability**<br>Operators may not receive a critical leak warning potentially leading to a hazard or safety risks. |
| Unauthorized Access | An attacker obtains a dashboard employee's credentials and logs into the system without permission. | Dashboard login credentials | **Confidentiality**<br>**Integrity**<br>**Availability** |

# Priority Ranking

Based on your analysis, rank the six attacks from **most dangerous** to **least dangerous** for The Grand Marina:

1.Unauthorized access

2.Man-in-the-Middle

3.Denial of service

4.Spoofing

5.Replay attack

6.Eavesdropping